CSTE4103 — Cloud Computing

Virtual Networks and Virtual Storage

Virtual Switches, VLAN and VXLAN, Overlays, SDN and OpenFlow, and Block, File, and Object Storage

Lecture 4 • Cybersecurity Engineering — 4th Year

Dr. Ruaa Ali Khamees • ICT Department
Polytechnic College for Engineering Specializations – Baghdad, Middle Technical University

LECTURE 4 • OVERVIEW

Learning Objectives

1. **Virtual networking**
   Explain how VMs connect through virtual NICs and virtual switches, and the main network modes.
2. **VLAN and VXLAN**
   Compare them, and explain why clouds need more than VLANs.
3. **Underlay and overlay**
   Describe how tunnels build virtual networks on top of a physical IP network.
4. **SDN and OpenFlow**
   Describe the control plane and data plane split, and how flow tables work.
5. **Storage types**
   Compare block, file, and object storage, and traditional versus software-defined storage.
6. **Security**
   Evaluate segmentation, encryption, and access risks in virtual networks and storage.

1 — WHY

From Cables to Software

Lecture 3 gave us thousands of VMs on a few hosts. Nobody can re-cable a data center each time a VM appears, so the network must be built in software too.

Isolation per tenant

Each customer needs a private network that other customers cannot see, even on the same physical switches.

Agility

A virtual network, subnet, or firewall rule is created in seconds by an API call, not by a technician with a cable.

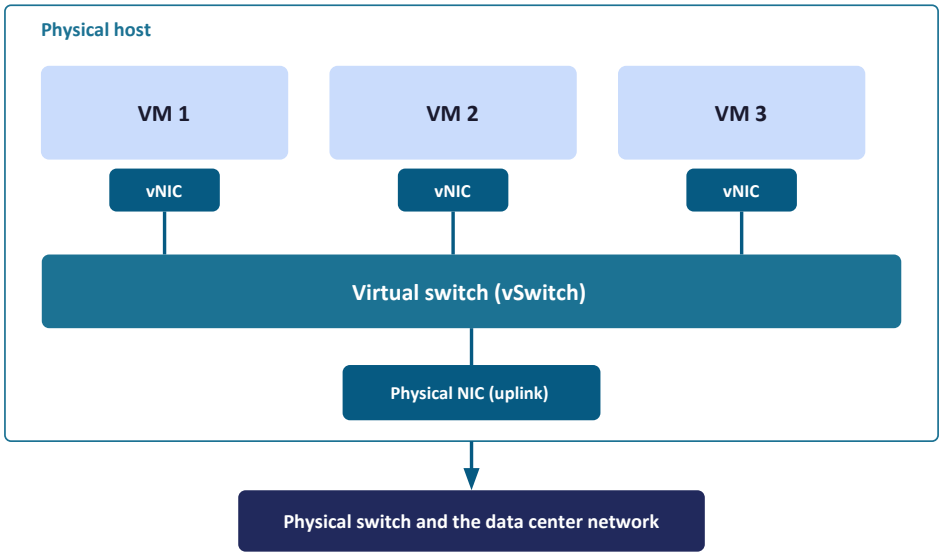
Mobility

When a VM live-migrates, it keeps its IP address and network policy. The network must follow the VM.

Link to Lab 3: the NAT network that gave your Ubuntu VM internet access was a virtual network created by VMware in software.

2 — VIRTUAL NETWORKS

Virtual NICs and Virtual Switches



How it works

- Each VM gets one or more virtual NICs with their own MAC address
- The vSwitch forwards frames between VMs on the same host in software
- Frames for other hosts leave through the physical uplink

Examples: VMware vSwitch, Hyper-V vSwitch, Linux bridge, Open vSwitch (OVS), Docker bridge.

⚠ Security angle: Traffic between two VMs on one vSwitch never meets the physical firewall. Use port security and limit promiscuous mode.

2 — VIRTUAL NETWORKS

Network Modes: What You Choose in Lab 3

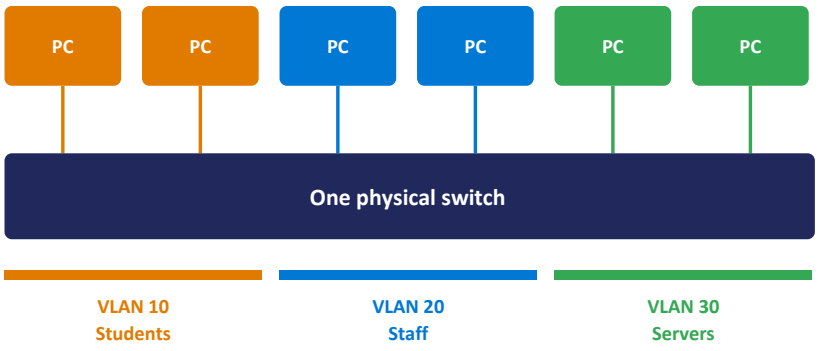
Mode	How the VM connects	Reaches the internet	Reachable from the LAN	Typical use
Bridged	Acts like a separate device on your real network with its own LAN address	Yes	Yes	Servers other machines must reach
NAT (used in Lab 3)	Shares the host's address; VMware translates addresses	Yes	No (unless port forwarding)	Labs, safe default for learning
Host-only	Private network between the VM and the host only	No	No	Isolated testing, malware analysis

Docker's default bridge network also uses NAT for outbound traffic.

⚠ Security angle: Choose the least exposed mode that works. Bridged puts the VM directly on your LAN.

3 — SEGMENTATION AND OVERLAYS

VLAN: One Switch, Many Networks



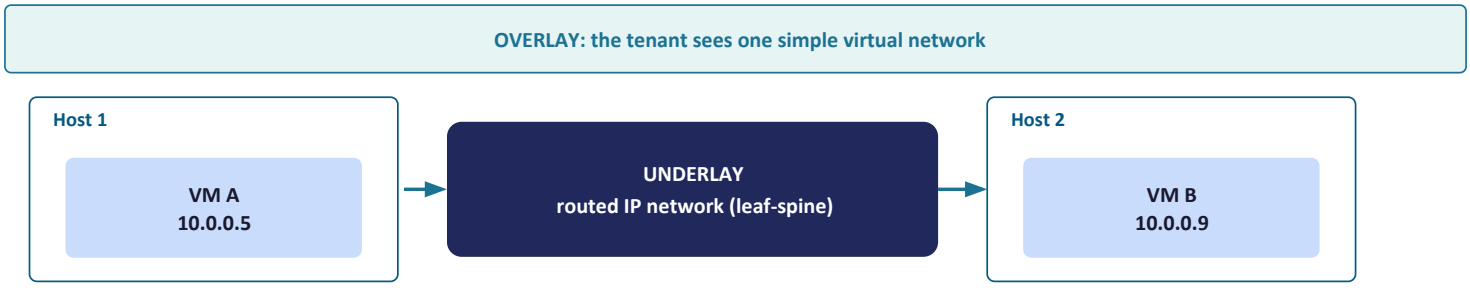
Same cables and switch, three separate broadcast domains.

How it works

- Access ports belong to one VLAN
- Trunk ports carry several VLANs, each frame tagged with 802.1Q (4 bytes)
- The VLAN ID has 12 bits: 1 to 4094 usable
- Different VLANs talk only through a router or Layer 3 switch

Why VLANs are not enough for the cloud: only 4094 IDs, and each VLAN is a flat Layer 2 domain tied to one switched network. A cloud with thousands of tenants needs a different tool.

Underlay and Overlay



1 Send

VM A sends a normal frame to VM B.

2 Wrap

The tunnel endpoint on Host 1 puts it inside an outer packet addressed to Host 2.

3 Route

The underlay routes the outer packet like any IP traffic.

4 Unwrap

Host 2 removes the outer header and delivers the frame to VM B.

Like a letter in an envelope: the postal network reads only the outer address. Two tenants can even reuse the same 10.0.0.0/24 range without conflict.

Wrap and Unwrap: The Four Steps

Step	Where	What happens	What is on the wire
1 Send	VM A, on Host 1	VM A sends a normal frame to VM B. It does not know a tunnel exists.	MAC A → MAC B 10.0.0.5 → 10.0.0.9
2 Wrap	VTEP on Host 1	Looks up (VNI 5001, MAC B) and finds Host 2. Adds outer Ethernet, IP, UDP 4789, and VXLAN headers (about 50 bytes).	192.168.1.11 → 192.168.1.12 UDP 4789, VNI 5001 + inner frame
3 Route	Underlay (leaf-spine)	Routes by the outer IP only, like any IP traffic. The UDP source port varies, so flows spread across the spines.	Switches see only host to host. Tenant addresses stay hidden.
4 Unwrap	VTEP on Host 2	UDP 4789 goes to the VTEP. It reads VNI 5001, removes the outer headers, and delivers by MAC B.	The original frame reaches VM B unchanged.

The reply from VM B takes the same path in reverse: Host 2 wraps, the underlay routes, Host 1 unwraps.

VXLAN: Layer 2 over Layer 3

VXLAN wraps the tenant's whole Ethernet frame inside a UDP packet, so a virtual Layer 2 network can span any routed IP fabric.

Added by the tunnel endpoint (VTEP): about 50 bytes of overhead



VNI: 24 bits

About 16 million virtual networks (2^24), against 4094 for VLAN.

VTEP

VXLAN Tunnel Endpoint. It sits in the host's vSwitch or in a leaf switch and wraps and unwraps frames.

MTU pitfall

The 50 extra bytes can push packets over 1500. Raise the underlay MTU (for example 9000) or lower the VM MTU to about 1450.

⚠ Security angle: VXLAN does not encrypt or authenticate. The VNI separates tenants logically only, so protect the underlay and use IPsec or WireGuard for sensitive traffic.

VLAN vs VXLAN

Aspect	VLAN	VXLAN
Segment ID	12 bits: 4094 usable	24-bit VNI: about 16 million
How it isolates	A tag on the Ethernet frame (802.1Q)	The frame is wrapped inside UDP/IP (tunnel)
Reach	Within one switched Layer 2 network	Across a routed IP network, even between sites
Overhead	4 bytes	About 50 bytes
Configuration	Switch ports and trunks, often manual	Tunnel endpoints plus a control plane (for example BGP EVPN), automated
Best for	Campus networks, small or fixed data centers	Multi-tenant clouds and large data centers

In practice they combine: VLANs inside a rack or campus, VXLAN across the data center fabric.

3 — SEGMENTATION AND OVERLAYS

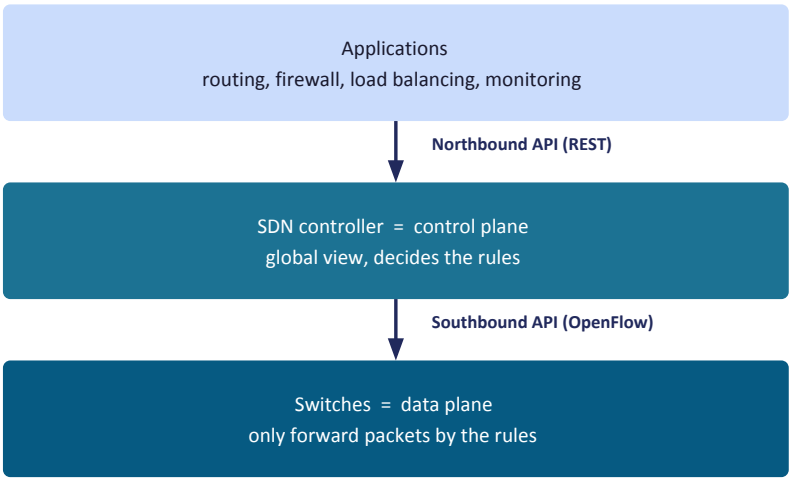
Virtual Networks in the Public Cloud

Building block	What it does	AWS	Azure
Virtual network	Your private address space in the cloud	VPC	VNet
Subnet	A slice of the network, public or private	Subnet	Subnet
Route table	Decides where traffic goes next	Route table	Route table (UDR)
Internet access	Inbound and outbound paths to the internet	Internet gateway, NAT gateway	Public IP, NAT Gateway
Firewall rules	Allow or deny traffic by port, protocol, and source	Security group (stateful), NACL (stateless)	NSG (Network Security Group)

⚠ Security angle: Default deny: open only what is needed. Never allow SSH (22) or RDP (3389) from 0.0.0.0/0. Keep databases in private subnets.

4 — SDN

Software-Defined Networking (SDN)



Traditional network

Every switch runs its own control logic and is configured one by one. Hard to change, hard to automate.

SDN

Logic is centralized in software. One policy change reaches every switch in seconds, through an API.

⚠ Security angle: The controller is a high-value target: whoever controls it controls the whole network. Isolate it, authenticate switches, encrypt the control channel (TLS), and restrict API access.

4 — SDN

OpenFlow: Match, Then Act

An OpenFlow switch keeps flow tables. Each entry says: if a packet matches these fields, perform this action.

Priority	Match	Action
300	src IP 10.0.1.5, TCP port 22	Drop
200	dst IP 10.0.2.0/24	Forward to port 3
100	VLAN 20	Forward to port 5
0	anything else (table-miss)	Send to controller

When there is no match

- Switch sends a packet-in message to the controller
- Controller decides and replies with a flow-mod
- Later packets of the flow match the new entry and are forwarded at full speed

In practice: Open vSwitch (OVS) is the most common OpenFlow software switch and is used inside many clouds. P4 goes further and lets you program how the switch itself parses packets.

5 — VIRTUAL STORAGE

Block, File, and Object Storage

	Block	File	Object
Data unit	Fixed-size blocks	Files in folders	Object + metadata + ID
Access	Raw disk: iSCSI, Fibre Channel	NFS, SMB over the network	HTTP REST API
Examples	AWS EBS, Azure Disk, SAN	AWS EFS, Azure Files, NAS	AWS S3, Azure Blob, MinIO
Strength	Low latency, high speed	Familiar, shared by many users	Huge scale, cheap, durable
Limit	One VM at a time, no sharing	Scaling and locking get harder	Not for databases or editing in place
Best for	VM disks, databases	Shared folders, home directories	Backups, media, logs, big data

5 — VIRTUAL STORAGE

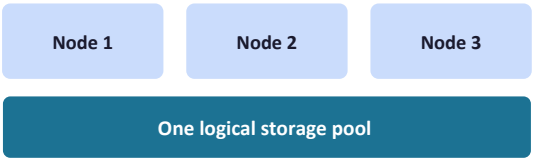
Traditional vs Software-Defined Storage

Traditional

- DAS: disks inside one server
- NAS: file server on the network
- SAN: dedicated storage network for block access
- Scale-up: buy a bigger, costlier array
- Vendor hardware and management tools

Software-defined

- Software runs on commodity servers
- Scale-out: add nodes, capacity and speed grow
- Pools many disks into one logical store
- Policies by API: replication, snapshots
- Examples: Ceph, HDFS, VMware vSAN, MinIO



Replication

3 full copies on different nodes. Simple and fast, but 3x the space.

Erasure coding

Data split into 4 pieces plus 2 parity: survives 2 failures at 1.5x space.

Replication protects against failure, not against deletion or ransomware: keep separate backups.

6 — SECURITY LENS

Security Lens: Networks and Storage

Area	Typical risk	Good practice
VLAN and vSwitch	VLAN hopping through trunk misconfiguration; east-west traffic that no firewall sees	Disable unused ports, fix native VLAN, micro-segmentation
Overlay (VXLAN)	Unencrypted tunnels; an attacker on the underlay can read or inject traffic	Secure the underlay, encrypt with IPsec or WireGuard
Cloud network rules	Security groups open to 0.0.0.0/0; databases exposed to the internet	Least privilege, private subnets, regular rule audits
SDN controller	Single point of control and failure; rogue switches	Isolate, TLS, authenticate, redundant controllers
Storage	Public buckets, unencrypted data, leaked snapshots and backups	Block public access, encrypt at rest and in transit, manage keys, versioning

7 — APPLY IT

Mini Case: A University Private Cloud

Scenario: a university runs a private cloud for three tenants: Teaching, Research, and Administration. Exam PDFs for 2,000 students, lab VM disks, and a shared Linux home folder must all be stored. Student VMs must never reach the administration network.

Q1 Storage type

Which type (block, file, object) for each of the three storage needs, and why?

Q2 Segmentation

How do you keep student VMs away from the administration network?

Q3 VLAN or VXLAN

Would VLANs be enough here? When would you move to VXLAN?

Q4 Threats

Name two threats to this design and one control for each.

SUMMARY

Key Takeaways

- A virtual switch connects VMs in software; the mode (Bridged, NAT, Host-only) sets exposure
- VLANs give 4094 segments at Layer 2; VXLAN gives about 16 million over routed IP
- Overlays ride on an underlay; they isolate logically but do not encrypt
- SDN separates control from data plane; the controller is a prime target
- Block, file, and object storage serve different needs; software-defined storage scales out

Next lecture

Distributed Systems and Cloud-Native Software: consensus and Raft, consistency and CAP, cloud-native vs conventional, MapReduce, Hadoop, HDFS

Midterm exam: 10 November, covering Lectures 1 to 4.

Reference: Comer, D. E. (2021). The Cloud Computing Book. CRC Press.